

Student number:



EEE3094S 2024: Class Test 1

- The test is open book. You do not need to hand in your resources.
- **The supporting information for all questions and formula sheet are attached at the end of this paper. You may remove them for easy access.**
- Answer on the test sheet. If you run out of space, you may attach additional pages, but please note that you have done so (e.g. write “see attached pages” by the question) and make sure they are marked with your student number.
- You may do your diagrams in pencil but please ensure that your lines are clear.
- There are 3 questions worth a total of 56 marks. The maximum mark you can score for the test is 50 though.
- You have 90 minutes to complete the test + 15 minutes of reading time.

Question 1

The aim of this question is to create a block diagram describing altitude control in the enhanced drone. It uses information from the 'Altitude Control of the Drone' section of the information sheet.

Draw a block diagram describing height control of the drone. Your diagram should include the following subsystems: (A large space is provided on the next page) [20 marks]

- The model-predictive controller, NMPC
- The flight controller, FC
- The motors (M)
- The propellers (P)
- The drone body, B
- The plastic shell, S
- The accelerometer (modelled as described)
- The tracking camera, TC
- The force/torque sensor, FTS

Indicate the following signals on your diagram:

- Goal position p_{ref}
- Position input to flight controller q
- Thrust input to drone F
- Gravity g
- Collision disturbance d
- Altitude y

(You may also include any other elements that you feel are necessary or helpful.)

Question 1

Question 2 [18]

This question develops an open-loop model of the basic drone. It uses information from the 'Open Loop Model of the Drone' section of the information sheet. Relevant plots are included in the 'Open Loop Responses of the Drone' section.

- The relationship between the input EMF $e(s)$ and thrust $F(t)$ produced by the motor-propeller subsystem can be approximated by a first-order model. Give this model as a transfer function and explain why it is an acceptable approximation of this system's behaviour. (Answer in terms of the given variables, rather than numerical values for the parameters.)
[5 marks]
- Following on from *a*, give an open-loop transfer function model for the drone relating $e(t)$ to the vertical **velocity** $\dot{y}(t)$, which incorporates the first-order approximation of the motor and propeller. (Answer in terms of the given variables, rather than numerical values for the parameters.) [2 marks]
- Your model in *b* should be a second-order system, but it is possible to simplify it even further, down to a first-order transfer function. Based on the given vertical velocity response of the drone to a unit step in the input $e(s)$, explain why this is an acceptable simplification. (Hint: are there any specifically second-order features in this step response?)

Suggest a suitable first-order model of the form $\frac{A}{Ts+1}$ **based on the step response**. You may annotate the copy of the plot to show important features that support your reasoning [4 marks]

- d) Consider the given Bode plot of the system. Is there a range of frequencies where you would say a first-order assumption holds? (Answer to the nearest order of magnitude/ multiple of 10). Explain. If so, use features of the response within this range to confirm the first-order model you obtained in part c. [4 marks]
- e) Does the first-order model you obtained from observing the system's step response and frequency response seem correct considering the measured parameters of the drone's motors, propeller and body? [3 marks]

Question 2

[illegible]

[illegible]

[illegible]

Question 3 [18]

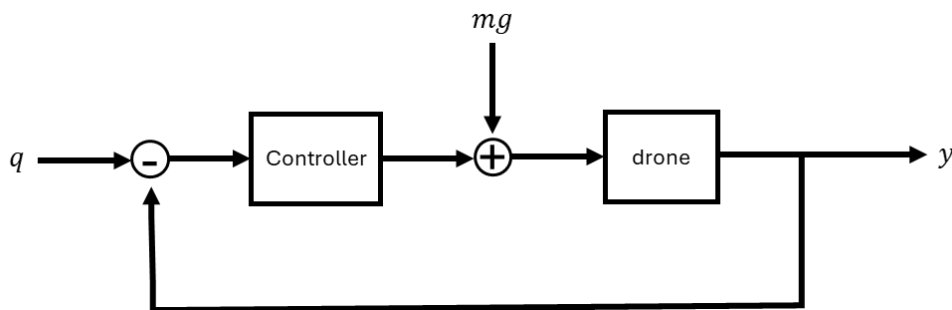
This question uses the drone model derived in the previous question to design a simple flight controller to regulate altitude. (Note: the altitude model is just the integral of the vertical velocity model you derived before.) If you had trouble getting a model, you can use the transfer functions below to represent the relationship between altitude and EMF:

$$\text{Full-order model: } G(s) = \frac{0.25}{s(0.00014s+1)(0.0013s+1)(2s+1)}$$

$$\text{Reduced-order approximation: } G_r(s) = \frac{0.25}{s(2s+1)}$$

Relevant plots are included in the 'Open Loop Responses of the Drone' section of the information sheet.

You want to design a simple flight controller that will receive the desired altitude input $q(s)$ from the NMPC and adjust the altitude of the drone to match it. You can assume that the drone is flying in open skies (no collisions) and that the sensing of the altitude is perfect, giving the simplified closed-loop system shown in the block diagram below.



- Sketch rough root locus plots for the drone, modelling the relationship between height y and EMF e using (i) the full fourth-order model, (ii) the reduced-order model. Hint: this is a *rough* plot, so your drawing does not need to be to scale. [6 marks]
- Comparing the root locus plot in (i) to the one in (ii), is there any unique information that you get from the full-order model that is lost when you move to the simplified one? (Hint: consider the stability and transient response characteristics of the closed-loop system) [2 marks]
- BONUS: Explain why the closed-loop system will track the altitude perfectly, without disturbance from gravity, regardless of gain. What would be required to achieve perfect tracking of the vertical velocity? [4 marks]
- If you want to use a proportional controller, how large would the gain need to be for the altitude to track a *velocity* setpoint with less than 10% error? [2 marks]
- Based on the given Nyquist plot of the open-loop altitude model's frequency response, would you say that it is possible to achieve velocity tracking with less than 10% error without making the system unstable? Explain. [4 marks]

[illegible]

[illegible]

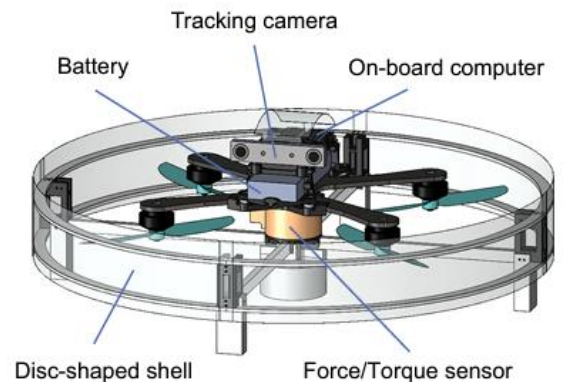
Information Sheet

All questions in this test are based on information in the following document. The content has been adapted and simplified to fit the context of a class test, but the ideas are consistent with real research published this year in the scientific journal 'Nature Communications'.

Background

For aerial robots (drones) to be useful in a variety of environments, they need to be able to navigate obstacles safely. Generally, this involves avoiding these obstacles entirely, but in the case of some compliant (bendy) objects such as leaves or soft branches and twigs, a better option might be to just push past them the way animals typically do.

One of the reasons this is easy for animals is because of their ability to sense collisions with objects anywhere on their bodies. Researchers are trying to give drones the same ability by surrounding them in bouncy plastic shells equipped with force/torque sensors (Aucone, et al. 2024)¹.



Altitude control of the drone (Question 1)

Control of the drone's position is accomplished in two stages: at the higher level, a nonlinear model-predictive controller (NMPC) uses a simulation of the drone's dynamics to determine the desired position of the drone, q . The lower-level flight controller (FC) converts this command to voltages applied to each of the drone's four brushless DC motors (M), causing it to rotate at the required velocity.

Each motor drives a propeller (P) causing thrust to be generated. The combination of these thrusts gives the total actuator force F acting on the drone's body (B), causing it to move to an altitude y . The body is also acted upon by gravity (g), and disturbances (d) caused by collisions with obstacles.

During vertical movement, it can be assumed that the drone body is parallel to the ground, with all motors receiving the same commands and turning at the same velocity.

The altitude and upward velocity ($\dot{y}$) of the drone are detected by two sensors:

1. An accelerometer, which can effectively be thought of as a reading of the *force acting on the drone's body*, scaled by $\frac{1}{m}$, where m is the drone's mass, and integrated.
2. A tracking camera (TC). The tracking camera gives a position approximation based on landmarks, which is then differentiated to estimate the velocity.

The force created by collision disturbances is sensed by a force/torque sensor (FTS), which works by measuring the deflection of the plastic shell (S) produced by d .

¹ Aucone, Emanuele, et al. "Synergistic morphology and feedback control for traversal of unknown compliant obstacles with aerial robots." *Nature Communications* 15.1 (2024): 2646.

The NMPC takes multiple inputs:

- 1) The ultimate goal position of the drone, p_{ref} .
- 2) The approximate position of the drone, which can be simplified as a weighted combination of the accelerometer and camera's position estimates. These are given weights w_1 and w_2 , respectively.
- 3) The approximate velocity of the drone, which can be simplified as a weighted combination of the accelerometer and camera's velocity estimates. These are given weights w_1 and w_2 , respectively.
- 4) The reading from the force/torque sensor.

Open-Loop model of the drone (Question 2)

The open-loop dynamics of the drone are a combination of the dynamics of the four motors and propellers, and the drone body. We will only consider vertical motion.

The input to the drone is the EMF $e(t)$ applied to the motors by the flight controller.

Motor Model

The relationship between torque input $\tau(t)$ and shaft velocity $\dot{\theta}(t)$ for each motor is given by the differential equation

$$J\ddot{\theta}(t) = -b\dot{\theta}(t) + \tau(t)$$

where J is the inertia of the motor, and b is a damping coefficient capturing the effect of friction.

The torque developed by the motor is related to the input current $i(t)$ by a scalar multiplier, the constant k_T .

The current is related to the input EMF by the transfer function

$$\frac{i(s)}{e(s)} = \frac{1}{R + sL} = \frac{\frac{1}{R}}{1 + s\frac{L}{R}}$$

where R and L are the resistance and inductance of the winding.

Note: it is important for friction in the motor to be very low for efficient operation. For this reason, the ratio $\frac{J}{b}$ will generally be very large compared to the electrical constant $\frac{L}{R}$ in a motor that is functioning properly.

Propeller Model

The conversion from motor shaft velocity to aerodynamic thrust performed by the propeller can be represented by a scalar, $k_p = \frac{F(t)}{\dot{\theta}(t)}$.

Drone Body Model

The upward velocity of the drone is related to the upward force applied by the differential equation

$$m\ddot{y}(t) = F(t) + a\dot{y}(t)$$

where m is the mass of the drone and a is a constant capturing the effect of aerodynamic drag.

Drone Parameters

Weighing the drone gives mass $m = 0.4$ kg

The datasheet of the drone's motors gives the following parameters:

Torque constant $k_T = 0.008$ Nm/A

Shaft inertia $J = 8 \times 10^{-6}$ kg.m²

Mechanical time constant $\frac{J}{b} = 1.3$ msec

Inductance $L = 1.8 \times 10^{-5}$ H

Resistance $R = 0.13$ Ω

From experiments, the propeller constant is estimated to be around $k_p = 0.0025$ Nm per rad/s

The aerodynamic drag constant is estimated to be around $a = 0.2$ N per m/s.

Open-Loop Response of Drone (Question 2,3)

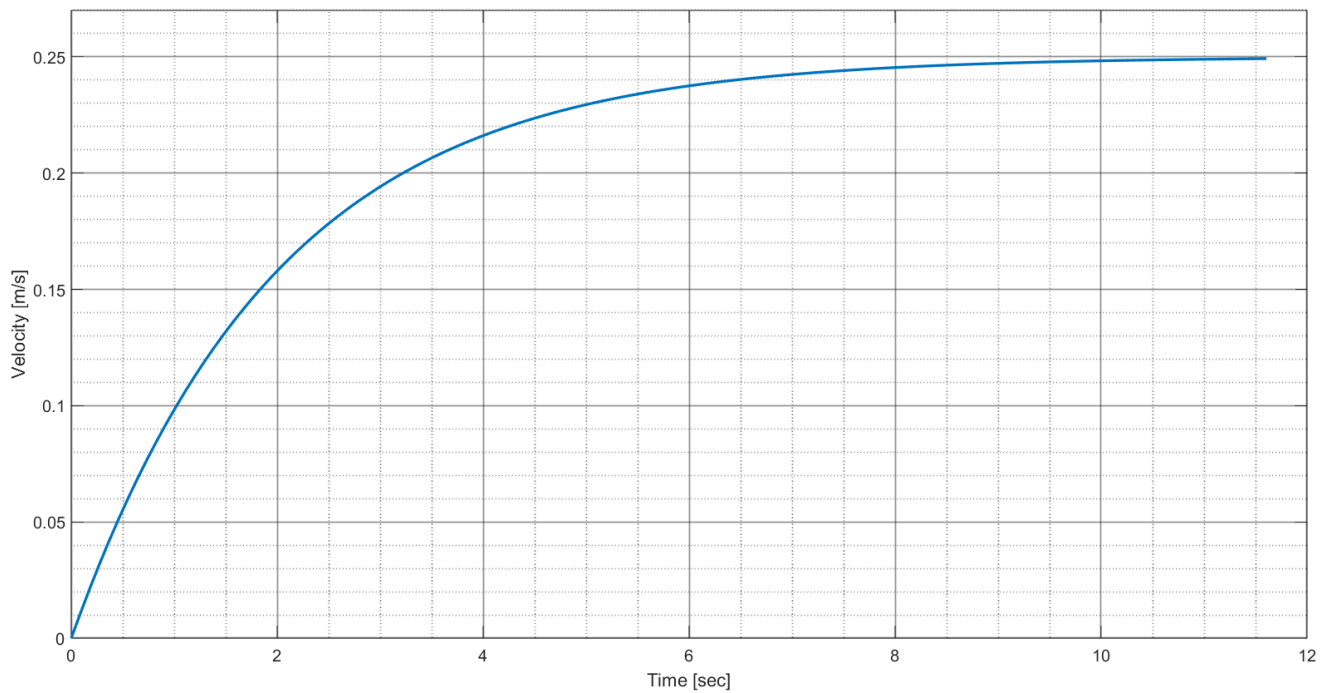


Figure 1: Vertical velocity of the drone in response to a unit step applied in the input EMF to the motors.

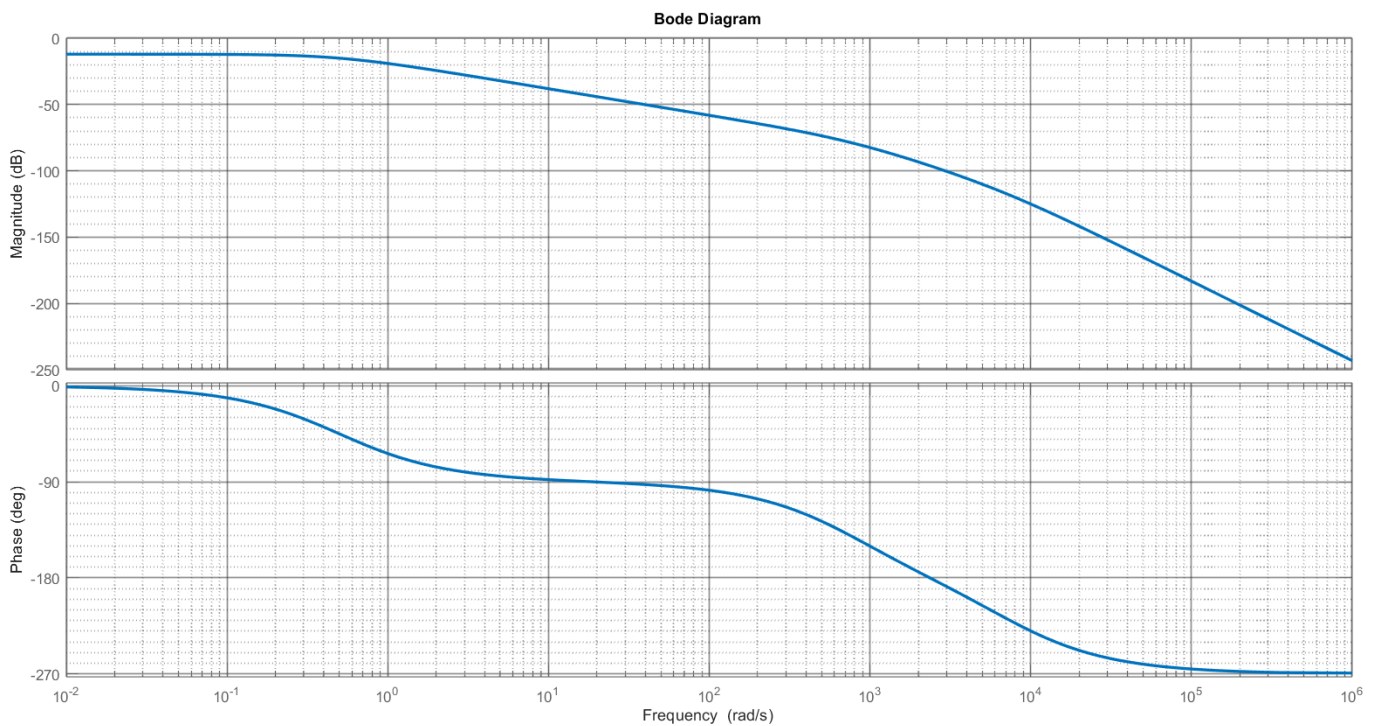


Figure 2: Open-loop frequency response of the drone's vertical velocity on a Bode plot

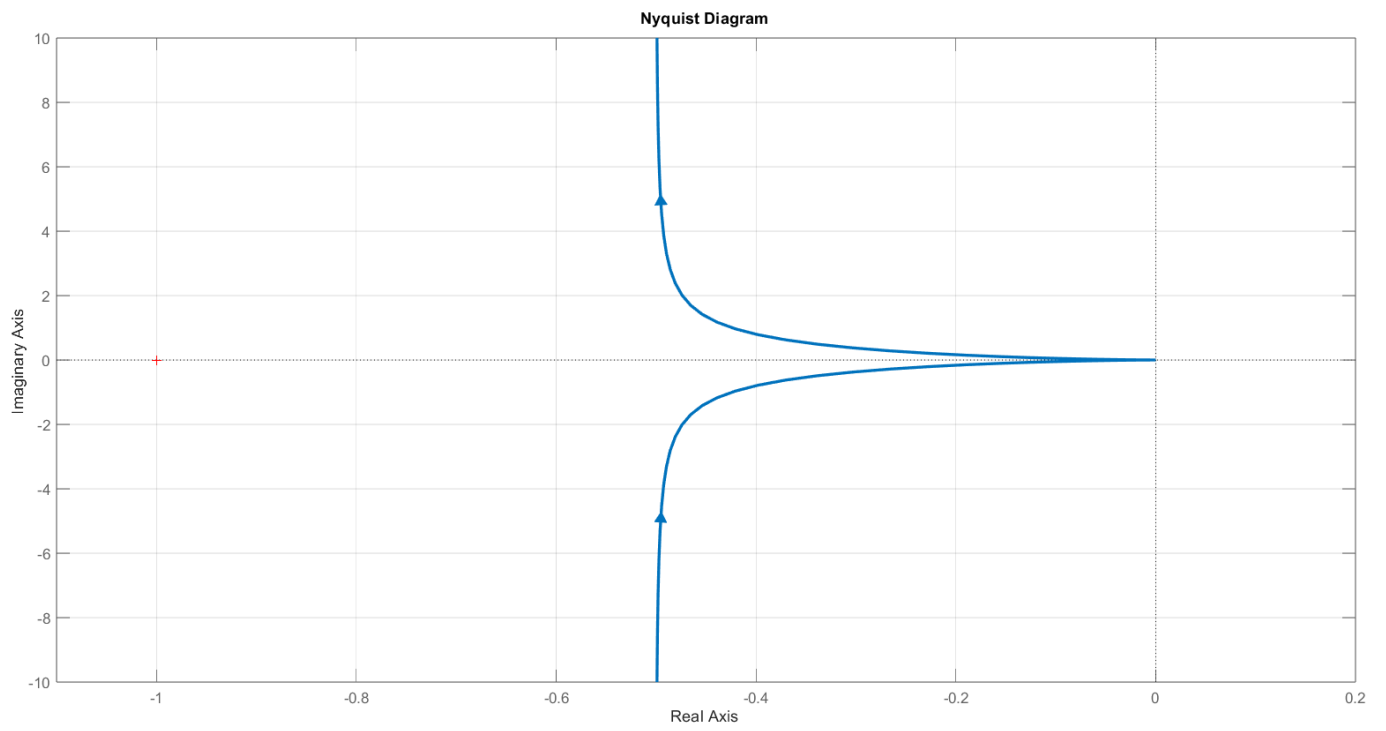


Figure 3: Open-loop frequency response of the drone's altitude on a Nyquist plot

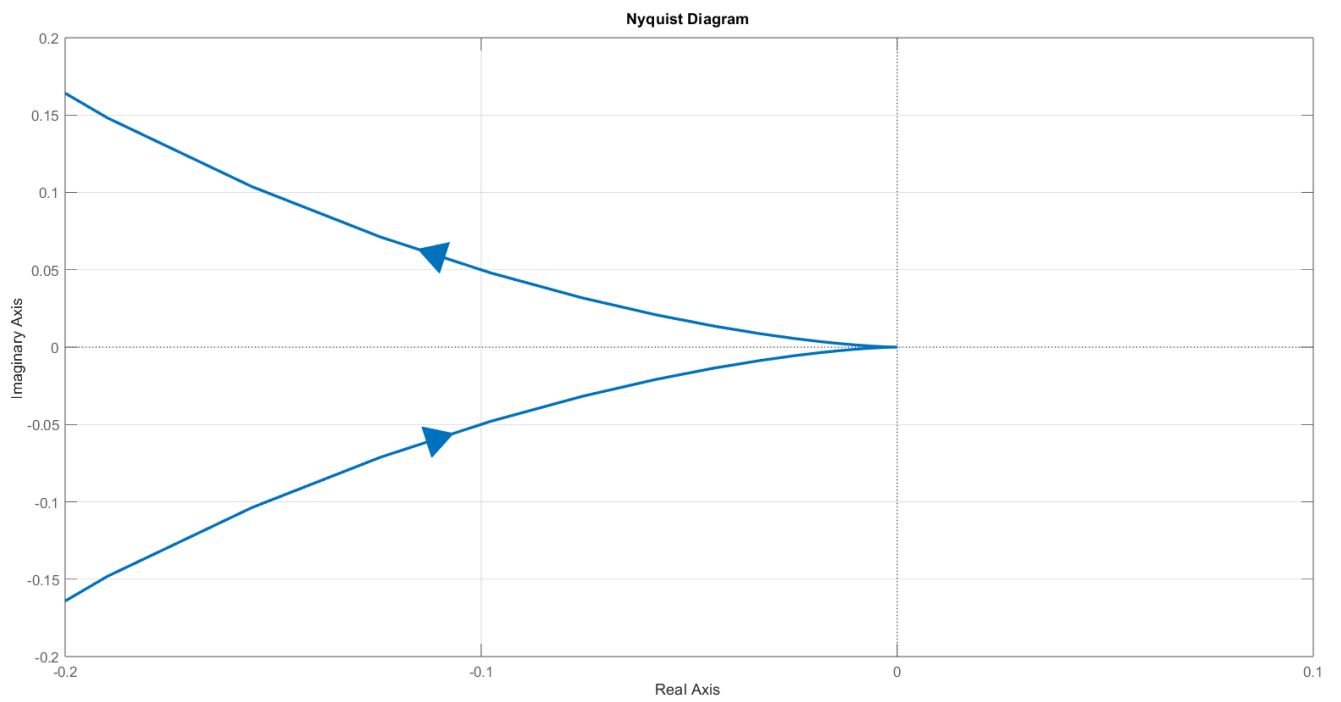


Figure 4: Detail view of the Nyquist plot in Figure 3.